

Digital Material Passport

ID DMP-2024-ARR-001 - Version 0.1.0  
Issue Date 2024-12-15 - Certificate Type EN 10204 3.1

| Manufacturer                                                                                                                                             | Customer                                                                                                                                                                  | Business Transaction |                       |
|----------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|-----------------------|
| Advanced Steel Technologies GmbH<br>Industriestraße 15<br>52070 Aachen<br>DE<br><a href="mailto:quality@advanced-steel.de">quality@advanced-steel.de</a> | Precision Engineering Ltd.<br>Manufacturing Park 42<br>Birmingham<br>B1 1AA<br>GB<br><a href="mailto:procurement@precision-eng.co.uk">procurement@precision-eng.co.uk</a> | Order                | PO-2024-7890          |
|                                                                                                                                                          |                                                                                                                                                                           | Delivery             | DN-2024-3456 - 500 kg |
|                                                                                                                                                          |                                                                                                                                                                           |                      |                       |
|                                                                                                                                                          |                                                                                                                                                                           |                      |                       |

Product

|              |                          |       |                                      |
|--------------|--------------------------|-------|--------------------------------------|
| Product Name | Hardenability Test Steel | Shape | RoundBar - D 100 mm - Length 1000 mm |
| Batch ID     | HTB-2024-045             |       |                                      |

Chemical Analysis

Heat Number H-2024-078 - Sample Location Ladle

| Symbol | C    | Mn   | Si   | P     | S     | Cr   | Ni   |
|--------|------|------|------|-------|-------|------|------|
| Unit   | %    | %    | %    | %     | %     | %    | %    |
| Min    | 0.42 | 0.5  |      |       |       |      |      |
| Max    | 0.5  | 0.8  | 0.4  | 0.03  | 0.025 | 0.25 | 0.25 |
| Actual | 0.45 | 0.65 | 0.25 | 0.015 | 0.008 | 0.12 | 0.08 |

Mechanical Properties

| Property                  | Symbol | Actual | Minimum | Maximum | Method  | Status |
|---------------------------|--------|--------|---------|---------|---------|--------|
| Jominy Hardenability Test |        |        |         |         | ISO 377 | ✓      |
| Quenched from 850°C       |        |        |         |         |         |        |

| Distance from - quenched end (mm) | 1.5  | 3    | 5    | 7    | 9    | 11   | 15   | 20   | 25    | 30    |
|-----------------------------------|------|------|------|------|------|------|------|------|-------|-------|
| Value [HRC]                       | 45   | 43   | 40   | 36   | 33   | 31   | 27   | 25   | 28    | 27    |
| Min                               | 42   | 39   | 35   | 32   | 29   | 26   | 22   | 20   |       |       |
| Max                               | 47   | 46   | 44   | 41   | 39   | 37   | 33   | 31   | <= 30 | <= 29 |
| Status                            | Pass | Pass | Pass | Pass | Pass | Pass | Pass | Pass | Pass  | Pass  |

Coating Thickness Profile

Magnetic induction method

ISO 2178 ✓

| Position along length (mm) | 0      | 50     | 100    |
|----------------------------|--------|--------|--------|
| Value [µm]                 | 125    | 115    | 108    |
| Min                        | >= 120 | >= 110 | >= 100 |
| Max                        | <= 130 | <= 125 | <= 120 |
| Status                     | Pass   | Pass   | Pass   |

Validation

We hereby certify that the material described above has been manufactured and tested in accordance with EN 10083-1 and ISO 377. All hardenability requirements have been met.

Validated by **Dr. Maria Schmidt**, Metallurgical Engineer,  
Quality Control Laboratory - 2024-12-15